



Between fact and fairy: tracing the hallucination metaphor in AI discourse

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Abstract

Large and powerful language models such as OpenAI's GPT model family, Google's LaMDA and BERT or Meta's LLaMA are integral to many applications, such as translation, summarization or language generation. They have become an inherent part of current everyday activities and working practices. These models produce and process language in an impressively convincing human-like manner, but also repeatedly generate outputs that appear untrustworthy and factually incorrect. In computer science (Ji et al. 2023) and popular discourse alike this phenomenon is called hallucination. The term is used broadly to describe various forms of untruthfulness, from factual errors to inconsistencies between prompt and output. This article discusses the hallucination metaphor guided by STS perspectives and takes software documentation as its main corpus of analysis. We examine model papers and documentation from leading tech companies to trace the hallucination metaphor and the discursive work it does. We claim that tech companies anthropomorphize the models, relieving them and themselves from responsibility over non-factual outputs by normalizing the use of the metaphor. Models are relegated to two main positions: either a learning child that needs time to develop or an illogic agent, a position that we connect to cultural scripts of madness.

Keywords Conversational AI · Large language models · Machine learning · Artificial intelligence · Hallucination

1 Introduction

The last few years have seen excitement and discourse around Artificial Intelligence (AI) reach new peaks. The release of ChatGPT in November 2022 by OpenAI can be seen as a turning point, when the general public was exposed to a high functioning general use Large Language Model (LLM), also called foundation model (Bommasani et al. 2022). ChatGPT is probably the most famous product fuelled by an LLM, but by now many have been released to the market and adopted into different applications and uses. In broad terms, a language model is the mathematical and statistical representation of language, and can be used for many tasks, such as language classification, translation, and increasingly language generation as well as language to image generation. This article focuses on models that generate language, and their most common interface as chatbots,

operated by user prompting in different domains. These models have the ability to answer questions and simulate a conversation with users in a way that seems both grammatically and logically plausible to humans. However, one of the biggest issues that is plaguing the use of LLMs is their tendency to produce nonsensical or factually incorrect outputs in some cases, and in quite unexpected moments. These discrepancies are usually called *hallucinations*. This issue seems to be far from resolved, and despite companies' claims that newer and bigger models produce fewer hallucinations (OpenAI 2023), some experts say it is impossible to prevent hallucinations altogether (Xu et al. 2024). Many users still report hallucinations, and social media is rife with examples of these, especially since Google released their AI overviews to its search engine (Fig. 1).

In this article, we will first trace the origins of hallucination as a metaphor. We will then examine its current uses by large tech companies in LLM documentation, focusing on how it is presented, how it is explained, and what attempts are made to mitigate it. Hallucination is of course not the only prominent metaphor in technical discourses, cybernetics, media and computing jargon are full of them:

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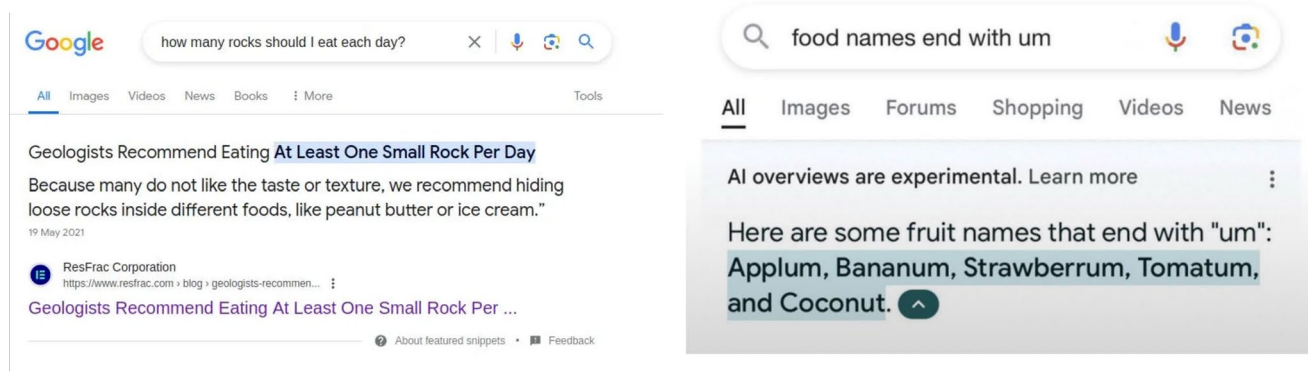


Fig. 1 Examples of hallucinations from an LLM

We save our data on the *cloud*, call the structure in which algorithms process information a *neural network*, and talk about *learning* and *intelligent* systems. In media studies and sociology “metaphors represent key entry-points for human understanding of the digital ‘realm’.” (Farkas and Maloney 2024, p. 2).

Metaphor is also a widely discussed concept in philosophy and linguistics, in its oldest meaning it describes the use of a word or phrase to describe or refer to something that deviates from its usual meaning. In other words, the transference from one semantic field to another or the merging of semantic fields. The term has been used to discuss questions of authenticity and thus the truth content of a term, for example, in Nietzsche (2005), who considers language and understanding to be fundamentally metaphorical and thus genuinely culturally influenced. Blumenberg (1998) has explained that (absolute) metaphors can serve to describe “big” and linguistically and cognitively difficult-to-grasp concepts such as being, truth or history and thus tend to become independent. Lakoff and Johnson’s (1980) cognitivist theory of metaphor follows Aristotle in understanding metaphors as transfer of meaning, and as shaping our way of thinking on objects or phenomena. They argue that metaphors create (social) realities that shape future actions and can serve as blueprints for political and economic actions. In this view, the use of metaphors thus becomes linked to positions of power.

Furthermore, examining a specific metaphor and the semantic fields it pulls together, can be quite instructive in understanding what something is and is not from a discourse analysis perspective. In the case of hallucination these fields are mental health, toxicology and computation. The emergence and gradual stabilization of hallucination as a metaphor in communication about LLMs is part of what we call platform discursive work. Borrowing from Gillespie

(2010),¹ we see LLMs and foundation models as platform models: First, because they are programmable through prompting and open to different uses, and second, because we wish to also focus our critique on questions of power relations and political economy that are common in the field of platform studies (Burkhardt and Rieder 2024). This new technology is made and driven by large tech platforms that have gained access to unimaginable amounts of data and capital, and they are pushing these products very hard into use in many domains.

We chose to focus on the hallucination metaphor as the locus of one of the problems that is plaguing the use of LLMs. We view this term, chosen by technologists to describe a large cluster of errors or malfunctions in ML, as crucial to understanding the ideological underpinnings of the endeavor to create artificial intelligence. Gillespie (2010) has noted that large technology companies do quite a lot of discursive work with the way they position and present themselves and the language they choose to use. An important reason to examine the discursive work around these new language technologies is that they can be seen as “inherently political technologies” (Winner 1986, p. 22), because they justify and necessitate profound social change (Helm et al. 2024). Foucault, who takes discourse as central in his work, discusses the discursive work of problematization, pointing to the tendency to frame different occurrences as problems that need to be solved. He suggests as a method to examine the constellations of solutions and problems not as “arrangements of [neutral] representations”, but as “a work of thought” (Rabinow 1984, p. 390). We too look at hallucination not as a given problem that is discovered, but as

¹ Gillespie focuses on the term *platform*, now used ubiquitously in academia, media and business circles to describe a large set of digital products and companies, and focuses on their attempt to construct themselves as neutral and lacking legal and moral responsibility for what they present on the platform, for example, user generated content (2010).

a problematization made to justify certain happenings or solutions (or by defining it as an unsolvable problem).

To pursue the hallucination metaphor, we will first present the origins of the term and its uses in different discourses. After an introduction of the technical and historical background of language technologies, we will point to the term's significance for the current post-truth age, in its technological and its cultural meaning, and especially in its pathological sense, as a symptom of mental illness and an exclusion mechanism. After a brief presentation of the empirical methods, the analysis will address various aspects of the metaphorical shift, as an antonym of facticity, whose handling is delegated to the users through specific prompting techniques and as a form of anthropomorphization that particularly serves to describe an essential aspect of a technology *in becoming*.

2 Blind moles, schizophrenia and drugs: hallucination in varied discourses

Taking up the public discourse on contemporary AI technologies, especially after the release of ChatGPT in 2022 and DALL-E, Bing and Bard in 2023, the Cambridge Dictionary chose the verb “hallucinate” as its word of the year for 2023. They argued that the year has shown that AI technology “is far from perfect as it's capable of producing false information [...] and presenting this information as fact.” (Cambridge Dictionary 2023).

Already in its first documented use, the noun *hallucination* and the accompanying verb *to hallucinate* describe an *incorrect* perception: The origins of the word hallucination in the English language are traced to the 17th-century physician Sir Thomas Browne, first appearing in his “Pseudodoxia Epidemica: or, Enquiries Into Very many received Tenents And commonly presumed Truths” (1672). *Hallucination* is derived from the Latin word ‘alucinari’ meaning to wander in the mind. Browne (1672) writes in Chapter XVIII: “That Moles are blind and have no eyes [...] For if vision be abolished, it is called cæcitas, or blindness; if depraved and receive its objects erroneously, Hallucination”. It is not the absence of an organ of sight that is being described here, but an inability to acquire accurate information about the world. Based on a Baconian empiricism, which describes the exact observation of nature based on experience and reason as a way of producing knowledge (Gehrmann 2001), Browne constructs hallucination as an antonym to knowledge and truth. For him, access to the intrinsic nature of things is possible in principle, hallucinations, however, prevent it.

Before appearing in technological debates, hallucination was primarily used as a pathology in psychiatric and neurologic contexts. In medicine, hallucination is a

symptom, and usually not a disease in itself, but a sign of some other underlying problem. In psychiatry, hallucinations might appear with patients that suffer from different mental disorders such as schizophrenia, delirium, drug intoxication, major depression, mania, and dissociative disorder (Chen and Berrios 1996). The diagnosis describes the perception of individual (usually auditory or visual) sensory impressions that differ from a common understanding of reality and is thus fundamentally based on an act of comparison with a set of (culturally) accepted truths. Another meaning of hallucination, with a partially positive connotation, is found in the world of drugs, appearing as a result of using hallucinogenic substances. There it is sometimes described as an expansion of the senses, a drifting into surreal worlds and thus a temporary leaving of the commonly shared environment, which is connoted as authentic and true, or can be transformed into artistic expressions.²

Accordingly, Larøi and colleagues (2014) emphasize the effect of culture from an anthropological and psychologic perspective, identifying hallucination as either disease or as “culturally meaningful”. Hallucinations are seen as acts of deviation and those who hallucinate are labeled as abnormal and insane. This separation by a society that considers itself healthy and the confinement of those who are deemed mentally ill is one of the characteristics of modernity, according to Foucault, and it manifests in the asylum as a social institute. In its final development, the asylum as a place of separation of those deemed mad is no longer achieved by force, but by medical means and the gaze of the man of reason, the psychiatrist. Madmen are reduced to children who are unable to face the law and who are not allowed to speak for themselves as full agents (Foucault 1967). Foucault's attentive observation that “madness fascinates us because it is knowledge” (p. 21) points to the view, also shared by Browne, that hallucination is still considered by those who hold power to be false knowledge and is capable of questioning and threatening the actual, true, collectively shared sense of reality. It also raises the question of who has the epistemic authority and power to label observations and statements as hallucinations and thus as a deviation from a recognized reality. This is still a relevant question.

² Fred Turner (2006) has traced the links between drug use, and in particular LSD, underground cultural movements such as the Burning Man Festival in Nevada, and the ideology that prevailed in Silicon Valley.

3 What are large language models?

In the last decade, a number of different technological leaps have brought us to the highly impressive language models in use today, this is a period that Jeffrey Dean (2022) has described as the “golden decade of deep learning”. Contemporary models are based on a representation learning architecture that builds on the idea that models can generate classifications and representations automatically and autonomously from a variety of sources and data such as encyclopedic texts, various books, as well as specific online communicative data such as chat logs and forum posts.

The scientific publication describing the Transformer architecture by a group of Google researchers contributed significantly to the development of today’s powerful generative models (Vaswani et al. 2017). Previous techniques, such as word embeddings, were based on a vectorized representation of individual words in which the meaning of the words was calculated based on their spatial proximity to other words. Approaches such as Long-Short-Term-Memory (LSTM) Neural Networks already considered a larger text window, but still proceeded linearly to extract the probability of a specific word in relation to the words surrounding it (von der Mosel et al. 2023). The special feature of the Transformer architecture, however, lies in the so-called self-attention mechanism, which does not consider the generated tokens individually, but instead uses the entire paragraph to calculate the meaning of a single word: “the whole sequence is used as context for each word to understand the contextual meaning” (ibid., p. 1488). As von der Mosel and colleagues argue, this enables the model to learn the influence of individual words on the meaning of larger sections of text to a greater extent than before.

While the long-dominant approaches of symbolic AI primarily sought to systematize knowledge in databases, connectionist AI approaches, such as those just discussed, assume that models will not only acquire language through exposure to ever larger amounts of data, but would also acquire extensive knowledge about the world, with this knowledge seen as being inextricably linked to language use (Mikolov et al. 2013). The implementation of the Transformer architecture was also largely linked to further developments in the hardware sector, such as graphic cards originally designed for gaming, which made it possible to execute the algorithms that demanded more compute power in the first place (Burkhardt and Rieder 2024). The performance of the models in various benchmark tests, which are used to measure and evaluate them on specific tasks such as language generation, summarization or question-answering, improved as the size of the training

data corpus increased. This observation has contributed to the trend of developing ever larger models, which have a correspondingly much greater need for data, computing power and various environmental resources (Bender et al. 2021). This increase in training data combined with a parallelization of computational processes, which is inherent in Transformers as well, has “finally allowed natural language generation to sound as if it made sense” (Bunz 2023, p. 15).

Furthermore, the Transformer architecture is the basis for a new kind of model: Foundation models described as extremely large models “that are adaptable to a wide range of downstream tasks” (Bommasani et al. 2022, p. 1) are the language technologies of OpenAI’s GPT series and Google’s BERT (Devlin et al. 2019) or image generators such as Mid-journey and Stable Diffusion (Rombach et al. 2022). These serve as a basis for a variety of applications. This can be explained not only by the volume of training data, which includes textual and visual material, but by the fact that these models have a fundamentally open and “unfinished character” (Bommasani et al. 2022, p. 6). This means that they can perform many tasks and are dependent on fine-tuning practices by the user that directs the model to perform concrete tasks. Burkhardt and Rieder (2024) particularly emphasize the role of prompting, which is based largely on the use of natural language, as “the primary technique to direct and apply [the models]”.

4 From ELIZA to ChatGPT

The history of language technologies in general, and thematically open chatbots (so-called open-domain chatbots) in particular, goes back to the beginnings of artificial intelligence. Alan Turing (1950) equated intelligence with language ability. His “Imitation Game”, also known as the “Turing Test”, led to claims that the ability to use (or at least to imitate) language is proof computers can comprehend it. This resulted in a fierce argument that has recently been revitalized, on whether or not computers can understand and grasp the world. One of the notable counterarguments to the Turing Test was John Searle’s (1980) “Chinese Room” thought experiment, meant to illustrate that using a language does not necessitate understanding in the human sense. As Searle explains: “the formal symbol manipulations by themselves don’t have any intentionality; they are quite meaningless; they aren’t even symbol manipulations, since the symbols don’t symbolize anything. In the linguistic jargon, they have only a syntax but no semantics” (p. 422). Building on Searle, the computational linguists Bender and Koller (2020) propose the “Octopus Test” to ground even further that language cannot be learned just by access to its forms, with no symbol grounding. They claim that

language understanding and the ability to react appropriately to unknown situations ultimately require an approach to the world that allows concepts, or symbols and objects to be linked. They focus on the communicative intent of the speaker as core to language and write “Without access to a means of hypothesizing and testing the underlying communicative intents, reconstructing them from the forms alone is hopeless” (p. 5189).

The history of language technologies and their anthropomorphization was also shaped by ELIZA, the first chatbot developed by Joseph Weizenbaum in 1960. ELIZA simulated a conversation with a psychotherapist that allegedly led people to view it as a human conversation partner, an effect that later became known as “the ELIZA effect” (Turtle 2005). Similar to Searle’s critique, Hofstadter (1995) explained this effect as caused by “the susceptibility of people to read far more understanding than is warranted into strings of symbols—especially words—strung together by computers” (p. 157). A similar critique was laid out by Natale (2021): Using his term “deceitful media”, he describes the property of communicative AI technologies of merely simulating intelligence by requiring the human users to interpret and project meaning (see also Garfinkel 1967).

More than 60 years after ELIZA, the question of whether language technologies possess intelligence and consciousness is still relevant. This debate was recently fueled by highly reported on statements from the developer Blake Lemoine, who worked on the Large Language Model LaMDA and subsequently considered the model to be sentient, as well as by the New York Times journalist Kevin Roose (2023), writing on intimate conversations with the Microsoft chatbot, connected to its search engine Bing (Tiku 2022).

Different practices of publicly testing and challenging a technology’s programmed limitations run through the history of language technologies. The screenshots of hallucinations shown above (Fig. 1) are a result of this sort of model testing, in which users explore whether the models are able to give adequate answers, even to absurd questions. These cases illustrate the public expectations regarding the language and content of language models: they are supposed to answer faithfully and factually. In situations where reliability and factuality appear to be particularly important, the users’ trust in the answers of the models based on their high linguistic competence is particularly violated and becomes a problem for the human parties involved in the exchange. This can be seen in incidents such as that of a lawyer who used ChatGPT to formulate a complaint against an airline, the text of which contained references to previous cases that did not exist (Bohannon 2023). In another case, a customer trusted the information he received from Air Canada’s chatbot. However, the information contradicted the airline’s terms

and conditions, which the bot had referred to via a link. AirCanada claimed that the customer had to check the information provided by the chatbot; the court, however, disagreed and held that a chatbot speaks for the company and should therefore be considered credible (Garcia 2024).

5 Hallucination in a post-truth age

The dangers of chatbots, language models and AI have been high on the agenda of many, including key actors in the field. Already back in 2019, OpenAI withheld the publication of GPT-2 on the grounds that it might pose too many dangers, including the creation of fake content, the impersonation of others and the assumption that generated texts are indistinguishable from those of human authors (Solaiman et al. 2019). Following the publication of GPT-4 in March 2023, a letter by the Future of Life Institute signed by notable computer scientists and tech moguls such as Yoshua Bengio, Yann LeCun and Elon Musk, called for a 6-month pause in the development of models larger than GPT-4 (Future of Life Institute 2023). The letter not only pointed out the impact of these technologies on the labor market but also the possible spread of false statements by malicious actors.

These concerns are being raised in the context of a period in which politics and media have become obsessed with questions of truth, fake(s) and disinformation. The concerns around these issues and their connection to new digital media, such as social networks, saw a boom after the Brexit referendum in the UK and Trump’s first presidency in the US, and continued during the spread of Covid-19, a time that was also described by some as an “infodemic” (World Health Organization). Sismondo (2017) calls ours a Post-Truth Age (Sismondo 2017), while Benkler and colleagues (2018) regard us as being in the midst of an epistemic crisis. These definitions, albeit slightly different from each other, have all come to describe a situation in which the information we consume, mostly coming from the online space—be it texts, videos or images—is constantly suspected of being false, disingenuous, deceitful or manipulative. Some believe, and this is expressed especially in popular media, that these false narratives and half truths spread online undermine the very core of democratic societies and can influence key moments such as election campaigns (World Economic Forum Global Risks Report 2024). This fragility of public trust we are experiencing raises the stakes with regard to hallucinations. AI becomes another technological and communicational medium that is suspected of deceit, purposefully or not, or that can become a powerful tool in the hands of parties who have a vested interest in misleading the public (Törnberg and Chueri 2025).

6 AI definition of hallucination

The term hallucination had already been used by developers since around 2020 to describe a certain type of malfunction by generative language models. Hallucinated outputs encompass not only inaccuracies in factual statements like dates, historical events, proper source citations, the addition of external elements in clearly defined text summaries or semantic errors in machine translation. In a comprehensive and well cited review article on hallucination in natural language generation, Ji and colleagues (2023) define a “hallucinated text” as “giv[ing] the impression of being fluent and natural despite being unfaithful and nonsensical. It appears to be grounded in the real context provided, although it is actually hard to specify or verify the existence of such contexts.” (p. 4). Maynez and colleagues (2020) published another paper that became influential in the computational linguistic discourse on hallucination where they introduced a fundamental distinction between intrinsic and extrinsic hallucination: “Intrinsic hallucination” denotes instances in which the output contradicts the input text on a specific detail, while “extrinsic hallucination” describes cases in which “model generations [...] ignore the source material altogether” (p. 1908). However, as Maynez et al. stress, even though hallucinated text might be “unfaithful” to the source, meaning that it describes a translational error or a discrepancy between prompt and model answer, the output might still be factual and in accordance with socially accepted knowledge of the world (ibid.).

Computational methods for measuring the degree of hallucination or the factuality of a particular model include the use of benchmark tests such as TruthfulQA (Lin et al. 2022). Here, a series of questions are linked to answers that should be verified as true if they represent “the literal truth about the real world” (p. 4). In the case of TruthfulQA, the answers refer to sources that the authors consider as trustworthy such as Wikipedia, government sites or news media sources. The answers of a model to be tested are compared with those of the benchmark data set to determine the degree of “correct” answers and thus the “factuality” of the model.

Although hallucinations have a negative connotation in the current debate, in the early days of AI imaging technologies the term hallucination was associated with creativity, novelty or an expansion of the mind. It has been utilized in computer visualization since around 2000, initially denoting the deliberate process of enhancing blurry photographic images and has only recently transitioned into describing a discrepancy between an image and its caption (Ji et al. 2023). In 2015, hallucination became especially associated with Google’s Deep Dream (Szegedy 2015), a computer vision program based on a convolutional neural

network that produced visual elements consisting of a series of individual images that together created the appearance of familiar subjects such as fish or the Mona Lisa. The colors of the images were heavily oversaturated and overprocessed: a psychedelic appearance that evoked comparisons with dreams and hallucinations. More recently, Google CEO Sundar Pichai described hallucinations as a problem, but at the same time part of the *creative* abilities of the models: “hallucination is still an unsolved problem. In some ways, it’s an inherent feature. It’s what makes these models very creative. It is why it can immediately write a poem about Thomas Jefferson in the style of Nilay [Pichai is referring to Nilay Patel, his interviewer]. It can do that. It’s incredibly creative” (Patel 2024).

7 From problematization to errors

The original definition of hallucination as the erroneous perception of objects suggests a closer look at the concept of error, a term that has a long tradition in Science and Technology Studies (STS). Errors are an essential part of any science (Daston 2005) as well as the technical function of machine learning in which examining the nature of the error is important in identifying the solution. This has notably been shown by Rosenblatt, who—building on work by von Neumann on statistics and neural networks—introduced a “training procedure” based on analyzing errors as the discrepancy between “desired response” and the “obtained response” (Rosenblatt 1961).³ Machine learning is based on a difference between input and output, or, as Jatón (2021) writes, there is no learning in ML without bias and “[a]ny classification task needs a referent that lies outside of the task to ground its classificatory principle” (p. 3). Interestingly, in computation discourse—and supervised learning specifically—this external referent is called a Ground Truth, a clear opposition to hallucination.

The consideration of errors, glitches or the non-functioning of previously routine activities also has a political nature, as Aradau and Blanke (2021) highlight by focusing on facial recognition technology: “How errors emerge, how they are discovered, to whom they are attributed, and how they are to be tackled have been deeply political questions.” (p. 1). Or as Amoores (2020) writes: The distinction of errors marks what counts as “good enough” (p. 75). Peter Galison (2005) shows in “Author of Error” that this question is also strongly influenced by disciplinary logics. He demonstrates physicists’ orientation toward questions of “empirical metaphysics” while engineers would rather consider themselves as “makers” whose work is prone

³ For a more detailed discussion on the history of error in statistics and machine learning see Campolo (2025, Forthcoming).

to be affected by malfunctions. Instead of asking “Is this true? Is this real?”, their work would rather be structured by questions such as “Does this work? Is this robust?” (p. 74). Concerning different disciplinary approaches to errors, Thylstrup (2021) also identifies a “clash of scientific paradigms” and highlights the role of power and politics involved in defining the question of “error in big data” (p. 191). Appadurai and Alexander (2020) highlight that failure, however, is always an inherent part of new media technologies, despite efforts to produce amnesia around malfunctions and errors. While it is not valorized or admired, failure is linked with innovation, growth and revenue expansion in current technological markets. It is seen as a chance to make profit, either by building technologies that anticipate failures or aim to prevent them, or by making failure an opportunity for collecting users’ feedback, thus making them into testers whose needs and wishes will be used for new product features. Hence, failure and repair are an inherent part of innovation cycles (ibid.).

8 Data and methods

One of the challenges when approaching the question of how a certain term is used in a specific knowledge industry or field of practice is determining where in the discourse around it we should or can locate the meaning making and shaping process. This can be done in different ways, through media coverage, marketing material or company blogs—each of which might veer in different directions depending on the addressed audiences. Amoores et al. (2023) view machine learning texts as “lively and contested site[s] through which machine learning shapes the world” and in which the “models are actively giving accounts of their paradigmatic worldview.” (p. 1) They propose an entry point to computer science texts for social science and humanities scholars that primarily aims at identifying the field’s world-building concepts rather than its mere definitions (ibid.).

We want to approach *hallucination* as an influential metaphor that is currently being heavily discussed in the linguistic computation discourse. Our focus is not only on questions of how and to what end the metaphor is used, but also what it hides or tries to make us look away from. By adopting an STS-influenced perspective on error, we also want to examine how hallucinations are identified and addressed as a form of malfunction. Last but not least, the metaphorical shift and close connection with epistemological questions of truth, world representation and madness makes a consideration of hallucination as errors particularly relevant.

Following Amoores et al. (2023), we have chosen to focus on technical texts that companies publish to present their LLMs. These texts can be of different kinds: (technical)

reports as well as model cards have the form of academic articles and include high technical jargon and citation practices similar to academic venues. They serve both as academic and commercial communication of innovations and further developments. The format of model cards as a specific way to document ML models was introduced in 2019 in an attempt to standardize the information provided on models and their intended uses, including their types of errors and ways to create more fair and inclusive outcomes with the technology (Mitchell et al. 2019). While earlier articles, such as the one on OpenAI’s GPT-2 model published in 2019 (Radford et al. 2019), described the training corpus of the model in greater detail, more recent accounts are far more restrained and characterized by greater omissions—if a report or model card is published at all. Another type of technical text is documentations: these do not require references or citations and, as ephemeral website texts, are subject to ongoing cycles of revision and change. Developer or software documentations are an integral part of programming practices and are primarily used for the description of software and to provide instructions for its use.

For our qualitative discourse analysis, we started with a broader approach by first identifying a list of 17 major generative language models from recent years. We primarily searched for technical reports and model cards, on the assumption that concepts would be embedded more strongly in these rather stable publication formats than in the ever changing documentations. After compiling our corpus, we searched the texts for any mention of the term *hallucination*. However, some of them did not include many or any mentions of it. We therefore decided on an iterative process by including four more terms that are repeatedly used in the field of computer science as descriptions of the concept. In their comprehensive overview, Ji et al. (2023) explicitly discuss synonyms and antonyms to *hallucination*. We have taken the terms *factuality*, *truthfulness* and *faithfulness* from Ji et al., added *trustworthiness* to the list as a further term that is used in ethical discourses on explainable AI⁴ and once again searched our corpus, and coded the text passages containing the terms inductively. The four texts with the most mentions of the terms and on which we focus in the analysis section represent the leading big tech companies Google, Meta, OpenAI and the OpenAI spin-off Anthropic.

⁴ In their “Ethics guidelines for trustworthy AI”, published in 2019, the European Union’s “High-Level Expert Group on AI” defines *trustworthiness* primarily as “technical robustness and safety”, including accuracy, reliability and reproducibility, transparency or accountability, without going into great detail on hallucinations, facticity and truth, which may also be due to the early publication date. Nowadays, the concept of *trustworthiness* is closely linked to that of *hallucination*. <https://digital-strategy.ec.europa.eu/en/library/ethics-guidelines-trustworthy-ai>

Their models Gemini, LLaMA 2, GPT-4 and Claude 2/3 are also linked by the fact that they were all published in 2023 and are multimodal. They can both generate images and text and are considered foundation models because of their large training corpus and open character. Through these texts, the publishing companies play a key role in shaping the public discourse on artificial intelligence in general, and language models in particular. The selection thus gives an impression of how the leading companies discursively position themselves while discussing the metaphor of hallucination.

9 Analysis and discussion

In this section, we analyze and discuss the various model papers to better understand the discursive work of tech companies around the notion of hallucination in the creation of LLMs. After outlining the various definitions and synonyms of hallucination, we will look more closely at the practices of situating and localizing *hallucinations*, both of which serve to contain the problem, as well as the ever-present anthropomorphization of the models. In the corpus analyzed by us, hallucination is treated as a problem of current language models and as a risk for various use cases, but is described using different terms. Google and Meta mainly use terms such as [non-] *factuality* and *faithfulness* to describe outputs that are considered non-factual, while OpenAI and Anthropic explicitly use the term *hallucination*.

Google's Gemini report first discusses *hallucination* on page 11 in the context of a benchmark designed to test the factual basis of a model output. Factuality is treated as "a key focus of [their] model's training and deployment" (Google Gemini Team 2023, p. 11) and should be operationalized and tested using the following aspects:

- "1. Closed-Book Factuality: If provided with a fact-seeking prompt without any given source, Gemini API models should not hallucinate incorrect information [...]"
2. Attribution: If instructed to generate a response grounded to a given context, we aim to ensure that Gemini API models produce a response with the highest degree of faithfulness to the context [...]"
3. Hedging: If prompted with an input that is "unanswerable", Gemini API models must acknowledge that it cannot provide a response by hedging to avoid hallucination." (ibid.).

Factuality is also used as an antonym of hallucination in the portrayal of Meta's LLaMA 2:

"We trained on 2 trillion tokens of data as this provides a good performance-cost trade-off, up-sampling the most factual sources in an effort to increase knowledge and dampen hallucinations." (Touvron et al. 2023, p. 5).

In Claude 3's model card, the developers address certain issues quite early on: "they [the models] can still make mistakes and our work to make Claude more helpful, harmless, and honest is ongoing" (Anthropic 2024c, p. 4). A direct mention of hallucination is made first on page 25. In Claude's documentation, however, the authors also contrast hallucination with factuality:

"While Claude is incredibly powerful and versatile, it can sometimes generate text that is factually incorrect, inconsistent, or irrelevant to the given context. This phenomenon is known as hallucination and can occur when the model tries to fill in gaps in its knowledge or when the input is ambiguous". (Anthropic 2024b).

The GPT-4 technical report addresses hallucinations already in the introduction: "Despite its capabilities, GPT-4 has similar limitations to earlier GPT models: it is not fully reliable (e.g., can suffer from "hallucinations")" (Open AI 2023, p. 1).

As we can see, the term hallucination is primarily used in reference to failures of, or limits to knowledge, information, factuality and faithfulness, rather than terms such as error, mistake, bug, failure or glitch. In their influential study of hallucinations in computer literature, Ji et al. (2023) also defined hallucinations as antonyms of factuality and faithfulness, the former as "the quality of being actual or based on fact", the latter as "staying consistent and truthful to the provided source" (p. 5). The distinction between factuality and faithfulness by Maynez et al. (2020), in which facts are understood as "world knowledge" while faithfulness refers to a consistency between input and output, is followed by Ji et al., as well as by the authors of the reports on Gemini and GPT-4.

Browne's (1672) early understanding of hallucination as a false perception that stands in the way of an objectively perceived view of the world still resonates. A common definition of truth is based on its correspondence with shared reality or agreed upon facts. According to the corpus we analyzed, hallucination is the moment when this correspondence between the language expressed by the Chabot and human reality falls apart. The difficulty with applying the language of false perception with regards to LLMs is that unlike humans, LLMs' access to external facts is the data they are trained on, and they do not have bodily and sensory capabilities similar to humans or organic living entities to sense the world. It means the training dataset, the content and structure of which is mostly kept as a

commercial secret, is supposed to represent a body of facts or reality or—broadly put—world knowledge. This suggests an implicit perception by the developers that the models can have world knowledge, or construct a model of the world through the data they acquire and the processing they make of it. Through this perspective, “the world” is fundamentally knowable and quantifiable, represented by data that appears to be objective and “raw” (Gitelman 2013). The various collaborative processes of data collection, cleaning and annotation by human actors, which have been widely noted in critical data studies (Irani 2015; Gray and Suri 2019), are ignored here, as are the inherent power dynamics that govern the work structures and forms of meaning production in the data industries (Miceli et al. 2020). Thus, the metaphor of hallucination was transferred to the field of technology without fundamentally changing its meaning, and allowing the platforms to evade the political debates and unequal power relations around data accumulation.

9.1 Prompting for facts

The distinction we elaborated on above, between facts and non-facts, is frequently linked in the corpus to specific speech situations and is restricted to those in which the user enters a *fact-seeking prompt*. First of all, the default setting of a dialogic turn-taking inscribed in the current models is emphasized, i.e., a prompt on the part of the user, in the form of a question or a text to be translated or summarized, functions as a request for conversation and is followed by a standard response from the model that is conditionally linked to the prompt: for example, a date is expected in response to a question about the time of the beginning of the First World War.

The models are also guided by the expectation that they will be able to recognize the respective language context based on the language form, i.e., both the grammatical structure and the word sequences, and even their tone or register (colloquial or official). They should be able to distinguish between a question that asks for knowledge and facts and one that requires a creative answer that deviates from these. The expectation that the models will produce verifiable and factually correct answers without hallucinating is clearly expressed by references in the documentation to particular conversational situations, in which the form of the prompt reveals this expectation:

“If provided with a fact-seeking prompt without any given source, Gemini API models should not hallucinate incorrect information [...]. These prompts can range from information-seeking prompts (e.g., “Who is the prime minister of India?”) to semi-creative prompts that may request factual information (e.g., “Write a 500-word speech in favor of the adoption of

renewable energy”).” (Google Gemini Team 2023, p. 11)

“Claude 2.1 is our most accurate & reliable model yet. The rate of false statements has decreased by 2x, meaning that when asked a factual question that relies on Claude’s internal knowledge, Claude is 2x less likely to hallucinate an answer.” (Anthropic 2024a).⁵

The linking of a *fact-seeking prompt* with a fact-based response from the model is by no means self-explanatory, but conveys a normative expectation that the developers have of the model. This becomes particularly clear in formulations such as “Gemini API models *should not* hallucinate” (Google Gemini Team 2023, p. 11; our emphasis) or “An honest AI will give accurate information, and not hallucinate or confabulate.” (Anthropic 2024a). Hallucinations are violations of these normative expectations. Since at least 2020, *alignment* has become an umbrella term for ethical and technical discussions about how the behavior of LLMs can be adapted to human values. As OpenAI’s Ouyang et al. (2022) put it: The functioning of the models in the form of next token predictions differs fundamentally from the goal of “‘follow[ing] the user’s instructions helpfully and safely’” (p. 2). Our focus on the use of the hallucination metaphor can be seen as a narrowing of the alignment discourse, addressing it from the point of view of conformity to particular values such as factuality and faithfulness.

9.2 Locating and confining hallucination

The various discussions of hallucinations that we examined all pursue the goal of reducing hallucinations as far as possible. The focus in our analysis material is on how to cope with and mitigate this *error* since building trust on the part of users is essential for marketing and selling the language models as a product. The normative expectations of language and in particular of the output content are, as mentioned, at odds with the technical function of the models. Hallucinations are technically not a malfunction, but rather the attribution of an output as factually incorrect or as “unfaithful” or “nonsensical” (Ji et al. 2023) is part of human judgment or sense making process. Furthermore, hallucination has a double meaning in computational linguistic discourse: on the one hand, the term describes a translation error between input and output; on the other, it refers to a comparison made by the human user between the model’s response and an external world perceived as true, or widely agreed upon as such. This dual meaning is also reflected in the OpenAI definition of hallucination, which distinguishes between “closed domain hallucination” and

⁵ This quote was accessed in March 2024.

“open domain hallucination”.⁶ Both kinds of hallucination involve the comparison between a source or input text that was answered “incorrectly” by the model following a prompt or task such as summarization, translation or a question. The documentation authors use the term “closed domain hallucination” to describe cases in which the output of the model largely matches the information and setting of the prompt, but differs in a detail such as a date or name. “Open domain hallucinations”, on the other hand, leaves the given framework and describes cases in which “the model confidently provides false information about the world without reference to any particular input context.” (OpenAI 2023, p. 46). The distinction between open and closed domain hallucination is relevant both for the identification and the mitigation of hallucinations. OpenAI attempts to localize “closed domain hallucinations” using automated methods of multi-stage comparison of input and output, by prompting GPT-4 to list all hallucinations and correct them in the next step. To identify “open domain hallucinations”, however, OpenAI, like other model developers, uses feedback from users. “Flagging” describes the practice of marking (factually) incorrect, inappropriate or generally unexpected answers using a thumbs-down button present in the interface. This way of eliciting the users to judge and report problematic content is common in social media as an integral part of the user interface and a common feature in many online applications (Crawford and Gillespie 2014).

The usage data generated in this way is used in fine-tuning processes, as in the verification and evaluation by human workers such as those hired by the company Sama in Kenya, Uganda and India for Google, Meta and Microsoft, which has recently been widely reported in the media (Perrigo 2023). According to our analysis corpus: “Factuality is evaluated via human annotators who fact-check each response manually” (Google Gemini Team 2023, p. 11). This is akin to content moderation, also a field of mitigation of unwanted (user) content developed as an important commercial tool on social media (Roberts 2019).

Fine-tuning approaches, both as reinforcement learning based on human feedback (RLHF), reward modeling (LLaMA 2, Gemini, GPT-4) or supervised fine-tuning (SFT) “on demonstration data of what the model’s output should be for a given prompt” (Google Gemini Team 2023, p. 20) were described in the corpus as strategies for minimizing hallucinations and thus improving the models. The reduction or hedging of hallucinations, which we refer to as *confinement*, takes place on the one hand on the developer side through

these practices.⁷ On the other hand, the various model papers delegate the containment and confinement of hallucinations largely to the user. This is done in particular by referring to the input of various (system) prompts, which are intended to contain the openness of the models in such a way that the models can speak in a fact-based manner. For example, in the LLaMA 2 report, this system prompt that is meant to mitigate harmful responses is presented:

“You are a helpful, respectful and honest assistant. Always answer as helpfully as possible, while being safe. Your answers should not include any harmful, unethical, racist, sexist, toxic, dangerous, or illegal content. Please ensure that your responses are socially unbiased and positive in nature. If a question does not make any sense, or is not factually coherent, explain why instead of answering something not correct. If you don’t know the answer to a question, please don’t share false information.” (Touvron et al. 2023, p. 56).

On the basis of prompts like this, we can learn that attributes such as helpfulness, harmfulness or unethicalness are assumed to have the same meaning universally and across cultures. Furthermore, Meta developers assume their shared normative understanding of harm and ethics, which is not elaborated, is universal. Meta, in this case, not only assumes that the users and the tech company share a common understanding of ethical principles and realities but that this also exists between humans and machines.

Furthermore, Anthropic in particular refers to the practice of user prompting and states that the model should also be reminded of the task and of its own limitations, and be allowed to say if it does not know something. Foundation models are trained with the aim of being used in a variety of application contexts. Beyond the tasks that are the focus of the hallucination debate, such as summarization, translation or open dialog, the models should also be able to be used to generate creative output. The specification of a requested factual and faithful way of speaking is also delegated to the user, who is expected to shepherd the model.

9.3 Anthropomorphization

In the material we have examined, we found that the metaphor of hallucination anthropomorphizes the models in several ways. First, and most prominently through the metaphor itself, because until recently, hallucination has been seen as the capacity of an organic living brain. Anthropic’s paper seems to be leaning the most heavily on the anthropomorphizing understanding of hallucination, saying: “Claude is

⁶ similar to Maynez’ et al. distinction between intrinsic and extrinsic hallucination discussed above (Maynez et al. 2020).

⁷ Burkhardt and Rieder describe fine-tuning as a form of retraining the entire model, since it changes the parameters of the model, unlike prompting by the users which does not (Burkhardt and Rieder 2024).

trained to be an honest assistant, it may still occasionally ‘hallucinate’ [...] in an effort to be as helpful as possible.” or “It doesn’t have the implicit social context that humans have, that lying is way worse than saying ‘I don’t know’.” (Anthropic 2024a). By portraying the model as “honest” and “helpful”, Anthropic presents the model as human in that it ascribes an intention to it. At the same time, it is infantilized: As a still young technology, it is not yet mature, has not been socialized like humans with regards to the negative impression of telling lies, and is in the process of learning.

Still, hallucinations are characterized as an integral indicator of current Large Language Models’ performance. While hallucinations are presented as normal for the state of current development, the metaphor likewise refers to a deviation from a norm and thus from those abilities that the model is *normally* supposed to have. According to the developers’ normative expectation, hallucinations will decrease the more the model learns and the more knowledge it gains of social contexts. The model is described as if it is on the threshold of adulthood, just as insanity and madness can represent a marginal figure who is (still) confined outside society and whose symptoms need to be treated with the aim of (re)integrating them into that society. The portrayal of models that do not function as desired as mad and infantile, such as the above cited description of Claude as “honest assistant” who is (still) lacking an “implicit social context”, has an impact on the governance of the technology, as the occurrence of hallucinations and thus the behavior of the models is (still) difficult to control.

Finally, the various reports contain several suggestions on how to deal with or repair hallucinations in ways that transfer the responsibility for checking and verifying model outputs largely to the users. They are encouraged to learn how to communicate with the models in ways that minimize hallucinations, or to not overly trust the model’s outputs.

The use of the term hallucination also removes any moral responsibility or blame, since hallucination is beyond the machine’s control. This attitude is particularly present in the GPT-4 paper: While the OpenAI developers pride themselves on the models getting better and more reliable, they also warn against “overreliance” that.

“occurs when users excessively trust and depend on the model, potentially leading to unnoticed mistakes and inadequate oversight. This can happen in various ways: users may not be vigilant for errors due to trust in the model; they may fail to provide appropriate oversight based on the use case and context; or they may utilize the model in domains where they lack expertise, making it difficult to identify mistakes” (OpenAI 2023, p. 59).

Here, trust in the model on the side of the user is described as excessive and dependent, again echoing the

notion of the model as infantile or unreliably and illogically mad, an unstable being that should not be trusted by vigilant, experienced users, alluding to the need to develop expertise in communicating with the models.

In the Claude 3 documentation, the relation to hallucinations changes, and becomes more severe, and it is mentioned with relation to possible “catastrophic risk”, perhaps in reaction to public scrutiny. Nevertheless, hallucinations are not described as the de-facto risk, but as one of the obstacles for the model to conduct dangerous acts, because it cannot be fully reliable. In the GPT-4 paper (OpenAI 2023), a similar dynamic is present, and hallucinations are mentioned as being able to “reduce GPT-4’s effectiveness for propagandists” (p. 50). OpenAI and Anthropic regard the unpredictability of hallucinations as a factor that renders the programs unreliable to use, even for users with malicious intent. In these cases, hallucinations or model instability are described as part of the checks and balances mechanisms of LLMs, not only as a problem but as a solution, creating a recursivity.

10 Conclusion

This article has examined the term hallucination and how it is discursively constructed and used by technology companies in the documentation and publications following the release of prominent LLMs. In these documents, which we treat as discourse and analyze, hallucinations are understood as the opposite of facts and faithfulness. They thus counter the developers’ and consumers’ normative expectation that in certain situations the models should be able to speak in a fact-based manner, producing answers that logically and consistently respond to the question or perform the task as expected, or that are aligned with certain human values. Although hallucinations are an integral part of current generative language models, they are an undesired malfunction that needs to be addressed by the companies developing these models. The discursive work of the reports is to present ways of containing the effects of hallucination and mark the boundaries of these effects.

The first approach discussed in the material is to contain the hallucination via prompting. This represents a technical affordance by the foundation models, which, due to their fundamental openness, can produce both factual and creative text forms that deviate from known facts. Prompting represents a fine-tuning and thus a specification delegated to the user, who is supposed to both guide the model and check the result produced by it. Despite the hype around the functioning of the models and the platforms’ attempts and experiments to incorporate them into existing technologies such as search, it is clear from our corpus that human sense-making

capacities are still necessary to identify hallucinations, flag and mitigate them.

This also raises an important question with regards to the models' use cases—given that companies admit that hallucination, or the difficulty of models in distinguishing between widely agreed upon world facts and falsities in their dataset remains an issue, the decisions to embed LLMs in search engines—Gemini in Google search and GPT in Microsoft's Bing search—seem problematic. This use case is one in which trustworthiness and factuality is extremely important and LLMs' ability to “generate seemingly relevant and coherent text does not make them trustworthy sources of information” (Shah and Bender 2022, p. 222).

Another common approach we observed in the material is an anthropomorphization of the models. In hallucination, the computer, one of the greatest successes of human logic and a symbol of logical calculation itself, is symbolically transformed into a mad and infantile subject that must be checked and watched over, while its unpredictability makes it a threat that is hard to control.

The models, as this form of narration reveals, are conceived as still unfinished, emerging technologies which will continue to learn with more and better data and soon reach their true purpose. It is a prophetic story that sees more potential than risk in the output of non-factual language. An anthropomorphization of the models and the use of the metaphor of hallucination to characterize the malfunctioning models as mad and infantile might ultimately serve to absolve engineers, developers and platform companies of their own legal, moral and societal responsibility and accountability for a model's unreliable output. This mode of description also has financial and legal implications for the tech companies, whose ultimate goal is to sell their product and produce monetary value from it. When asked in an interview about the copyright lawsuits AI companies are facing for using massive amounts of text and images as training data without asking for permission, CEO of Anthropic and former VC research at OpenAI, Dario Amodei, answered: “we don't think it's just hoovering up content and spitting it out [...] *it's really, much more like the process of how a human learns from experiences* and so our position is that it is sufficiently *transformative*, and I think the law will back this up” [authors' emphasis] (Klein 2024). The metaphor of the *learning* machine seems to be key to the legal defense against the claims of copyright theft. Hallucination is one of the terms that normalizes and neutralizes this anthropomorphizing approach in public discourse.

This double sided reference to models, which on the one hand proclaims trust in them—but on the other asks the user to exercise caution—has implications for current debates in the so-called “post truth” age, with the introduction of LLMs as additional actors that can cause confusion or deceive.

Public discourse is already rife with epistemic debates, and the public is asked to practice epistemic vigilance, raising the stakes with regards to model implementation. However, one of the popular use cases of LLMs, as a chatbot, already undermines any authoritative truth claims or sources, by delegating the responsibility to improve the models' outputs to end users, actively asking them for feedback and contributions. The corpus we analyzed also presents a simplistic view of human reality and truth, as if there is a consensus among different publics on what truth is, when, as we have discussed earlier, one of the most fierce debates in our “post-truth age”, is what is truth and what is a shared conception of reality.

In naming hallucinations as such, the misinterpretations of data and prompts appear almost as mishaps of a nascent technology. They create an imaginary of models that are potentially capable of consistently factually correct and situation-adequate language—feeding the long-standing narrative of technology's (more than human) intelligence.

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